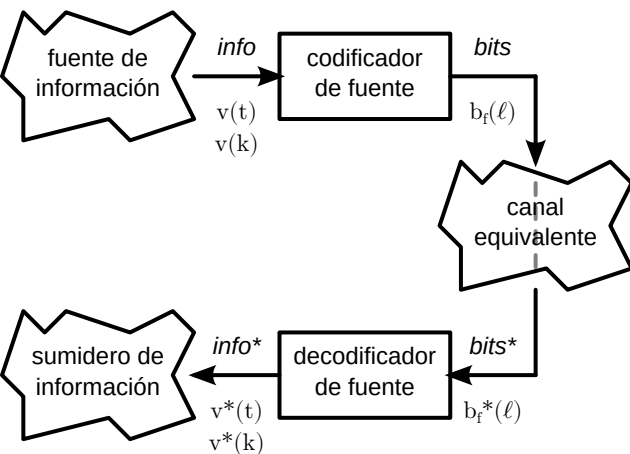
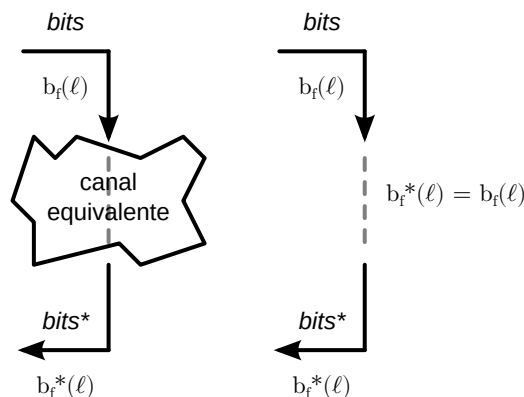


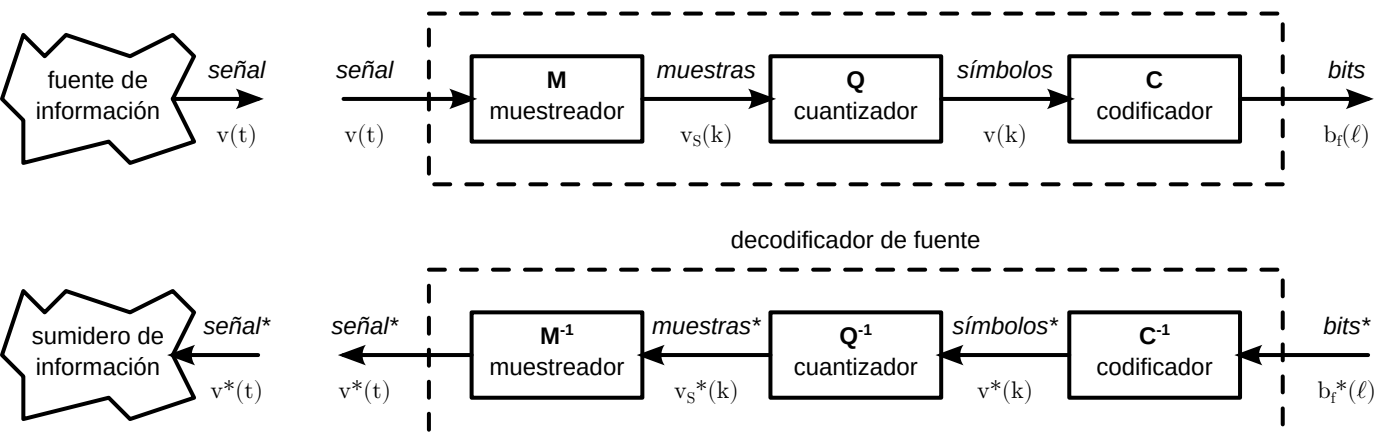
CODIFICACIÓN DE FUENTE



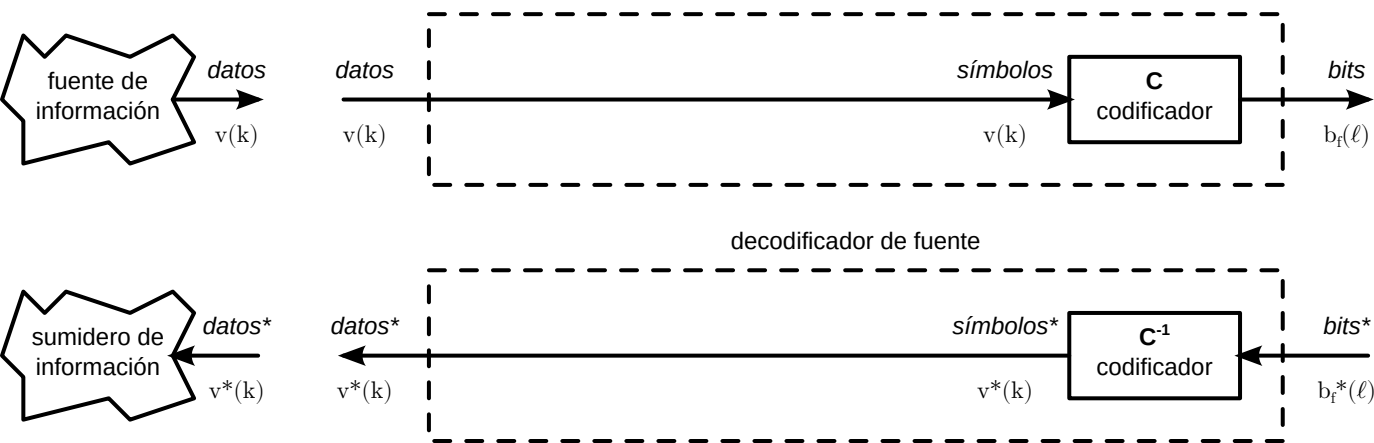
canal equivalente



info = señal



info = datos



M

Muestreador (regular)

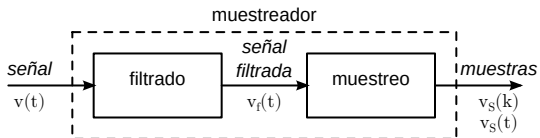
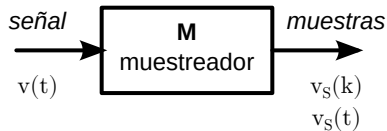
ancho de banda del filtro B_S [Hz] $B_S = B_v$

frecuencia de muestreo $F_S = 1/T_S$ [Hz] $F_S > 2 \cdot B_v$

$$v_S(k) = v_f(kT_S)$$

$$v_S(t) = \sum v_f(kT_S) \cdot \delta(t - kT_S)$$

$$V_S(f) = 1/F_S \cdot \sum V_f(f - kF_S)$$

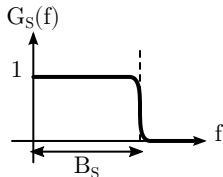


Filtro de muestreo (pasa-bajos)

$$g_S(t) \quad G_S(f)$$

$$v_f(t) = v(t) * g_S(t)$$

$$V_f(f) = V(f) \cdot G_S(f)$$



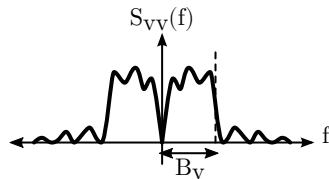
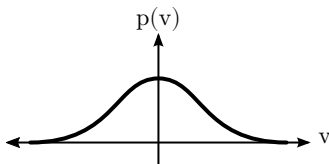
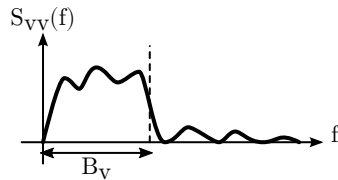
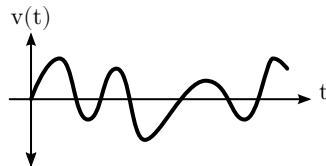
Señal de información

$v(t)$ $V(f)$

proceso estocástico continuo
señal aleatoria (de potencia)

ancho de banda B_v [Hz]

densidad de
probabilidad $p(v)$

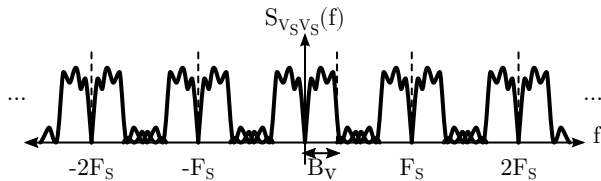
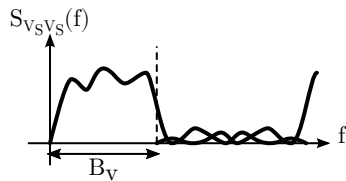
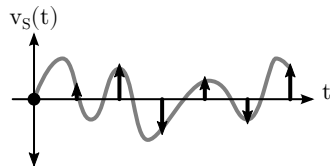


Señal muestreada

$v_S(t)$ $V_S(f)$

proceso estocástico particular
señal muestreada aleatoria

¿ancho de banda? B_V [Hz]



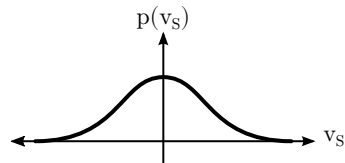
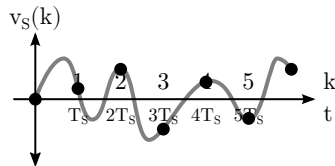
Secuencia de muestras

$v_S(k)$

proceso estocástico de tiempo discreto
señal aleatoria de tiempo discreto

tasa de muestras $D_S = 1/T_S$ [muestra/s]

densidad de
probabilidad $p(v_S)$



Q

Cuantizador (uniforme)

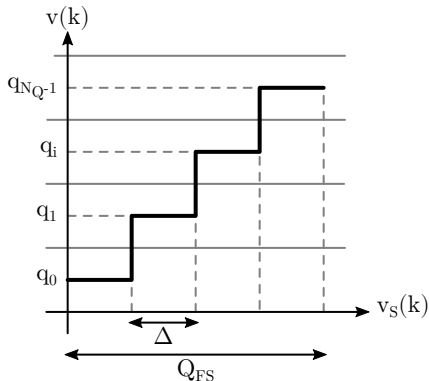
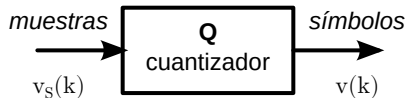
Resolución (cantidad N_Q
de niveles)

rango de plena
escala Q_{FS}

niveles $q_0, q_1, \dots, q_i, \dots, q_{N_Q-1}$

rango de cada
nivel $\Delta \quad \Delta = Q_{FS} / N_Q$

$$v(k) = Q(v_s(k))$$



Datos de información

$v(k)$

proceso estocástico discreto

fuente discreta sin memoria

tasa de símbolos $D_s = 1/T_s$ [símbolo/s]

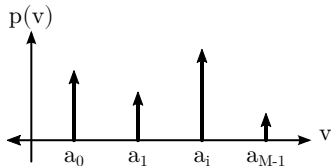
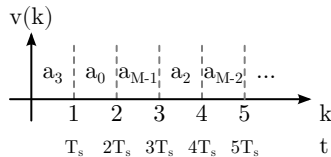
alfabeto $A = \{a_0, a_1, \dots, a_i, \dots, a_{M-1}\}$

tamaño del alfabeto M

densidad de
probabilidad $p(v)$

$v(k) = [a_3, a_0, a_{M-1}, a_2, a_{M-2}, \dots]$

$k = 0, 1, 2, 3, 4, \dots$



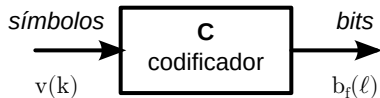
C

Codificador (binario)

cantidad de códigos N_C $N_C = M$

bits por símbolo b [bit/símbolo]

palabras de código $c_0, c_1, \dots, c_i, \dots, c_{N_C-1}$



símbolos	palabras de código	bits (b=3)
a_0	c_0	000
a_1	c_1	001
...	...	...
a_i	c_i	100
...	...	...
a_{M-1}	c_{N_C-1}	110

Bits de información

$b_f(\ell)$

proceso estocástico discreto binario

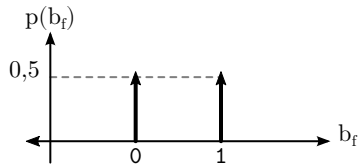
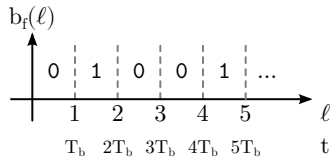
fuelle discreta sin memoria

tasa de datos $D = 1/T_b$ [bit/s]

probabilidad
de aparición $p(b_f)$
 $p(0) = p(1) = 0,5$

$b_f(\ell) = [0, 1, 0, 1, 0, \dots]$

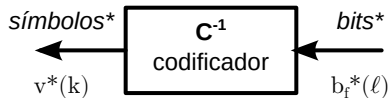
$\ell = 0, 1, 2, 3, 4, \dots$



Descodificador (binario)

bits por símbolo b [bit/símbolo]

palabras de código $c_0, c_1, \dots, c_i, \dots, c_{N_C-1}$

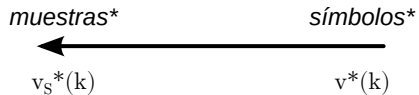
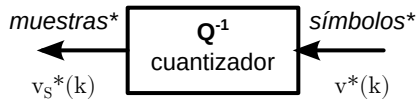


bits* (b=3)	palabras de código	símbolos*
000	c_0	a_0^*
001	c_1	a_1^*
...	...	...
100	c_i	a_i^*
...	...	...
110	c_{N_C-1}	a_{M-1}^*

Q^{-1}

Descuantizador (uniforme)

$$v_S^*(k) = v^*(k)$$



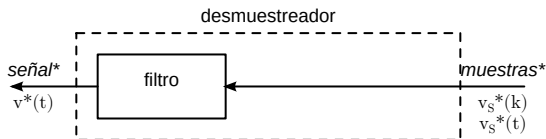
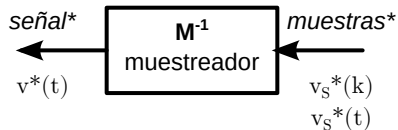
M^{-1}

Desmuestreador (regular)

ancho de banda B_S [Hz] $B_S = B_v$
del filtro

$$v(t) = v_S^*(t) * g_S(t)$$

$$V(f) = V_S^*(f) \cdot G_S(f)$$



Filtro de muestreo (pasa-bajos)

$g_S(t)$ $G_S(f)$

